

An Evaluation of 2022 Offensive Linemen in Pass Protection

My goal was to assess offensive linemen in the first 9 weeks of the 2022 NFL regular season through tracking data and play-by-play data, creating metrics to differentiate between high-performing and replacement-performing linemen. Using statistical methods and data science, I established a clear separation between OLs in performance and consistency, focusing on players' abilities to diffuse pass rushers, keeping them at a distance from the QB handling the ball, and delaying or preventing pressure on the quarterback. Elite linemen, for instance, should not only boast lower allowed pressure rates, they should also require significantly more time on average before allowing a pressure. In the end, I was able to develop metrics to divide the linemen into performance groups, create a simple scoring system (PPS) to rank linemen, and even search for potential value in players with less play time than frequent players performing at the same approximate level.

Using 9 weeks of the 2022 NFL Regular Season, I manipulated tracking data on the play level, aggregating them into metrics for each lineman and play. This included one for average separation between the player each lineman is blocking and the ball-carrier, who in this case was always the quarterback, time before pressure arrives, and how often pressure gets to the quarterback through each lineman. Once I'd finished with the play-level, I summarized each player's performance across the nine weeks I had. Cushion was the metric I focused the most on, and combined the distance between the pass rusher and qb with the duration the blocker held that distance, so a guard holding his ground on a long 5 second play would have a "thicker cushion" for that play than holding his ground for a quick slant.

In this plot, you can see how the average cushion for an elite lineman would be higher than that of a low-performing OL, accordingly with the higher average distance and longer average play length. Please note that these two players are the extremes of the PPS scale for demonstration purposes. PPS was developed as an unbiased score to rate the linemen in the dataset, rewarding a player for high average cushion and average time to allow pressure, but penalizing them for a high allowed pressure rate. Players the upper echelon included Tyler Linderbaum, Chris

Lindstrom, and Duane Brown, with rookies Kenyon Green and Luke Goedeke joining Saahdiq Charles in the bottom 3. Beyond this simple ranking, I wanted to divide players into performance groups.

By clustering on cushion, pressure rates, and successful blocking duration, four clear groupings formed. The first two are very similar, and include great players, with good reliability – you expect them to perform within a pretty narrow cone, and at a high level. The first contains the elite players, like Jason Kelce and Lane Johnson, and the second is just below that. Then you have a highly reliable group of middle-performing linemen, followed by an extremely volatile group. Not only are they inconsistent, they also perform at a poor level in comparison to the other groups, with high pressure rates and the lowest mean cushion values. This identifies our replacement level players.

Offensive linemen are a unique analysis challenge, because the stats aren't as clear cut as yards after catch or touchdowns, which is why in-depth analysis of this sort can be so beneficial. As just part of its many benefits, this can be used to identify which players to move on from, as well as players to have a deeper look into. This is just a small dive into this analysis, but I find it fascinating, especially if extended into analyzing run-blocking efficiency.

